# Peak-dBA Noise Model — Methodology & Measurement Record

**Purpose.** This document specifies the single-event peak-dBA noise model behind the TrueNoise sensitivity map, which compares today's BWI Runway 15R departure corridor against the proposed WARYN-BRGHT track, per home and per school. It also records the measurements the project rests on — the loudness law, the turn-altitude record, and the instrument cross-checks — and, at the end, clears up the technical points that are most often misread. It is written so another analyst can reproduce every figure.

*Prepared 2026-10-08; revised 2026-10-19. This revision adds the measurement basis, the 2,889-flight turn-altitude record, the five-flight calibrated-meter cross-check, the receptor characterization, equipment notes, a plain-language explanation of the loudness law, and a section clearing up common misconceptions. The model itself (the sensitivity-map figures) is unchanged from the 2026-10-10 version.*

### Measurement basis — the short version

TrueNoise is a **continuous measurement record**, not a model fed by assumptions.

Between **1 May and 15 August 2026** the monitor logged roughly **30,000 departure samples across 153 days**, grouped into about 3,185 departure passes. From that record:

- **2,889 clean Runway 15R departures** were analyzed for turn altitude — and the result held steady every month (median 2,325 ft in May, 2,375 in June, 2,292 in July, 2,196 in August), so it reflects a full season, not a snapshot.
- **249 isolated overflights** calibrated the loudness-vs-distance law, which then reproduced an independent calibration point (73.3 dBA at 0.55 mi) and the monitor's own measured level (70.5 dBA) *without being tuned to either*.
- A **five-flight blind test against a calibrated sound-level meter** put the model within about 0.8 dB of ground truth — slightly better than a commercial noise-tracking device measured in the same window.

Each claim below is tied to its own sample so the evidence can be checked: the altitude finding rests on ~2,889 flights, the loudness law on 249 measured overflights plus three independent validations, and the flight geometry on the ADS-B track set. The large altitude sample and the smaller law sample reinforce each other — the altitude the model assumes is independently confirmed on nearly 3,000 flights.

## 1. What the model does, in one paragraph

For every residential address in the corridor, and for named schools, the model computes the **single-event peak A-weighted sound level (peak dBA)** — the loudest instant of one departure overflight — once for today's 15R procedure and once for the proposed WARYN-BRGHT procedure, and reports both levels and their difference.

Peak dBA is the quantity TrueNoise argues matters (startle, speech interference, sleep disturbance) and the quantity the monitor measures directly. It is **not** DNL (a 24-hour energy average) and **not** $L_{Aeq,16h}$; those require daily operation counts and night-weighting and are deliberately out of scope (see §2 and the misconceptions section).

The mapped corridor follows **both** paths being compared — today's **west-curving** 15R track and the proposed **southeast** WARYN-BRGHT track — so the homes made quieter and the homes that would see an increase are both shown. A home is included where its modeled peak reaches **62 dBA under at least one** scenario: **33,206 residential address points**.

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## 2. What we measure, and why it is not DNL

Almost every misunderstanding of this work starts here, so it is worth being exact. There are two different ways to put a number on aircraft noise:

**Single-event peak dBA** — what TrueNoise measures — is the loudest instant of *one* overflight: the spike you actually hear when a plane passes. It is an event, measured directly by the monitor.

**DNL** (Day-Night average sound Level) is a 24-hour *energy average* of all noise, quiet hours included, with a 10-dB penalty added to night events. It is a bookkeeping figure, not a sound anyone hears, and the number "65" that appears in federal land-use guidance is a DNL contour.

These are different physical quantities on different scales. A corridor can sit far below DNL 65 as a 24-hour average and still deliver single-event peaks in the low 70s dozens of times a day — which is exactly the Severn situation. For that reason the figures in this paper **must not be compared to, or converted into, a DNL 65 figure**, and must not be fed into DNL-based annoyance or reading-score formulas; those require the averaged metrics and a separate computation. The model answers one question — *how loud is each pass, where* — and nothing about 24-hour averages.

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## 3. Inputs (all measured or from public records)

| Input | Source | Description |
|---|---|---|
| Departure tracks | `bwi_tracks.db` | ADS-B `track_point` rows; 90 clean 15R departures after filtering. Source of the measured climb and the measured west turn geometry. |
| Turn-altitude record | `aircraft_log_2026-05…08.csv` | 153 daily calibrated noise logs, ~30,000 departure samples → 2,889 clean 15R overflights with altitude at closest approach (§6). |
| Corridor homes | Anne Arundel County address points | "Structure Address Points," residential building codes, swept to the 62 dBA peak corridor of either track → 33,206 homes. |
| Noise event logs | `aircraft_log_2026-*.csv` | Per-event measurements (dBA, slant range, altitude) from the TrueNoise IEC 61672 Class 2 monitor. Used to fit and validate the loudness law. |

No modeled or default manufacturer noise tables (ANP/AEDT) are used; the loudness law is fit directly from the TrueNoise measurements.

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## 4. The loudness-vs-distance law — in plain terms, then exactly

**In plain terms.** Sound gets quieter with distance in a regular way, and the project measured that falloff directly. The rule the measurements produced: **every halving of the distance between you and the aircraft adds about 7 dB, and every doubling subtracts about 7 dB.** Anchor that to one measured point — about **73 dBA when the plane's closest approach is 0.55 mi away** — and the loudness at any other distance follows:

| Closest approach (straight-line) | Modeled peak | Everyday comparison |
|---|---|---|
| 0.25 mi | ~81 dBA | Garbage disposal; diesel truck passing close |
| 0.55 mi | ~73 dBA | The project calibration point (measured 73.3) |
| 1.0 mi | ~66.5 dBA | Busy traffic from the sidewalk |
| 2.0 mi | ~59 dBA | Normal conversation |
| 4.0 mi | ~52 dBA | Light rain; a quiet street |

**Exactly.** Peak level falls with slant range (the 3-D straight-line distance from the home to the aircraft at its closest point — horizontal offset and altitude combined) as:

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peak_dBA = A + B × log10(slant_range_mi)
A = 66.54   B = -24.37
```

**A (66.54)** is the intercept — the modeled peak at exactly one mile, where the log term is zero. It is the anchor of the curve, not a threshold or a standard. **B (−24.37)** is the slope — how fast loudness falls with distance (the "about 7 dB per halving" above; equivalently an energy coefficient K = −B/20 = **1.22**). Slant range is floored at 0.10 mi so the log stays finite directly beneath the path.

**How it was fit.** From the measured event logs, the single loudest sample of each pass was taken — **249 overflights** — and peak dBA was regressed against log10(slant). That regression gives A and B.

### Validation — three independent checks

**(1) An untouched calibration point.** The fitted law lands at 72.9 dBA at 0.55 mi against the project's measured record of 73.3 — a 0.4 dB agreement it was not tuned to hit.

**(2) End-to-end, against the monitor's own exposure.** With the measured climb in place, the full model reproduces the receptor's measured level: **70.6 dBA modeled vs 70.5 dBA measured** (median of 249 logged overflights at that location). The climb and geometry were not tuned to this number — it falls out of the independently-measured climb and the independently-fit law — so the agreement validates the climb curve and the law *jointly*. This check is what exposed an earlier contaminated climb (which had read the receptor ~4 dB low) and confirmed the fix.

**(3) A live, blind test against a calibrated meter** — described next.

## 5. Live cross-check against a calibrated meter 5 FLIGHTS · BLIND

On 15 October 2026, five consecutive 15R departures over 8312 Banister Rd, Severn (~130 ft from the monitor) were measured three independent ways at once: the **model** (computed from the ADS-B slant range *before the logs were opened*), a commercial **airnoise.io** device, and the **calibrated IEC 61672 Class 2 meter** taken as ground truth.

| # | Aircraft | Slant (mi) | Model peak | airnoise peak | Calibrated (truth) | Model – cal |
|---|---|---|---|---|---|---|
| 1 | 737-800 | 0.58 | 72.3 | 71.3 | 72.8 | −0.5 |
| 2 | A320 | 0.85 | 68.2 | 66.5 | 67.4 | +0.8 |
| 3 | 737-800 | 0.83 | 68.5 | 67.0 | 67.3 | +1.2 |
| 4 | 737-800 | 0.66 | 71.0 | 68.9 | 70.2 | +0.8 |
| 5 | A300F | 0.97 | 66.8 | 66.7 | 66.6 | +0.2 |

Against the calibrated meter, the **model's error was +0.50 dB mean, 0.76 dB RMS, 1.2 dB worst case**; the commercial device ran −0.78 dB mean, 0.98 dB RMS. Both instruments landed **well inside the ±7 dB screening band**, the model slightly ahead of the commercial device, and the two bracket the truth (the model runs a touch hot, the device a touch low; their average sits ~0.3 dB from the meter). *Caveat:* n = 5, a single 10-minute window and one weather state, with no near-overhead pass, so the steepest geometry and the slant floor are not exercised here. It is a confirmation, not a second calibration.

## 6. How high the aircraft actually are: the turn-altitude record 2,889 FLIGHTS

The single most-contested point in community discussion is how high aircraft are when they turn. It has a measured answer, on a large sample, and the answer is that **there is no single turn altitude** — the turn is a process, not a point.

Across **2,889 clean 15R departures (May–August 2026)**, altitude at closest approach to the receptor has a **median of 2,325 ft** (middle half 2,141–2,617 ft; only 7.8% below 2,000 ft, 0.6% below 1,700 ft). Closest approach falls at a median 3.46 nm from the airport — the turn-completion / over-community zone. Interpolated to a fixed 3.05 nm, the median is 2,047 ft. The figure is stable month to month:

| Month (2026) | Median completion altitude |
|---|---|
| May | 2,325 ft |
| June | 2,375 ft |
| July | 2,292 ft |
| August | 2,196 ft |

Putting the geometry together: departures **begin the turn around 1,800 ft** (near 2 nm, from the ADS-B tracks), **complete it around a median 2,300 ft** (2.5–3 nm out, over the community), and keep climbing to the west.

Because they are still low early, the aircraft are **lowest over the close-in neighborhoods and schools nearest the runway** (Quarterfield, and the communities just south of the field) and **higher by the western end of the turn**. Quoting any one number for "the turn" — whether a low figure near initiation or the ~2,300 ft completion median — is what produces the confusion in community debate; the honest statement names both the initiation and completion altitudes and the fact that the close-in homes see the lower end.

This also sets the stakes for proposals that turn aircraft *lower*: on the measured falloff, dropping the turn from ~2,300 ft to ~1,000 ft adds on the order of 6 dB to the peak directly beneath it.

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## 7. Core modeling choices

**7.1 Metric = single-event peak dBA** (§2). DNL and $L_{Aeq,16h}$ are deliberately excluded.

**7.2 Both procedures represented as idealized single lines.** WARYN-BRGHT is notional (unflown), so it has no measured dispersion; its design centerline is the honest representation of intent. For a *fair* comparison, today's 15R is also a single line (its median centerline), not its measured spread — comparing a single line against a dispersed cloud is invalid (the cloud has many chances to pass near each home) and is avoided.

**7.3 Today's 15R line = the ~79% west-curve majority, built along-track.** Measured 15R departures are bimodal: ~79% curve west, ~21% continue southeast. The "today" line is the median centerline of the west-curving majority, resampled by distance *along* the track (a straight-line-from-runway binning mis-behaves where the path curves). Past the measured turn it is extended along the published leg to the BOBYJ fix, which the measured track mass confirms to within ~0.1 mi.

**7.4 WARYN-BRGHT line** shares the runway segment (liftoff → WARYN identical to today by construction, so near-in differences are ~0), then runs WARYN → BRGHT and continues along the BRGHT design path.

**7.5 Altitude = the measured 15R climb, draped along-track.** Aircraft height is the measured median climb of the clean 15R departures, binned by along-track distance from liftoff (0.5-mi bins) from in-climb points only (excluding higher cruise traffic). It rises from ~1,800 ft at 2 mi to ~2,750 ft at 4 mi and ~4,500 ft by 8 mi. This climb curve is the piece independently confirmed by the 2,889-flight altitude record (§6) and by the receptor validation (§4).

**7.6 Peak dBA = the law evaluated at the 3-D minimum slant range** between each receptor and the line — the loudest moment of the pass.

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## 8. Uncertainty

**Error band = ±7 dB (95%).** The residual standard deviation of the 249-point fit is 3.7 dB; the 95% band is ±1.96·3.7 ≈ ±7 dB, reflecting real variation across aircraft type, weight and weather. **Differences smaller than this band are not individually distinguishable per home.** The defensible reading is the *spatial pattern* of where exposure shifts, not any single home's exact value — with the exception that the receptor's own modeled level is anchored to measurement (§4). (Modeling aircraft type explicitly tightens this band to about ±6.3 dB; that refinement is in progress and is not used in the map figures here.)

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## 9. Results

**Corridor (33,206 homes), single-event peak dBA.** Counts by band, order `<55 / 55–60 / 60–65 / 65–70 / ≥70` dBA:

- **Today's 15R departures:** 879 / 3,492 / 10,019 / 14,512 / 4,304
- **WARYN-BRGHT:** 4,501 / 5,317 / 11,627 / 7,210 / 4,551

By per-home change (WARYN-BRGHT minus today): quieter by ≥1.5 dB **16,772** · within ±1.5 dB **7,256** · louder by ≥1.5 dB **9,178**; **median change −1.75 dB**. The proposal is net relief by roughly two-to-one, at the cost of a real increase on the southeast side.

**Schools (peak dBA, ±7 dB), most-exposed-today first:**

| School | Today | WARYN-BRGHT | Δ |
|---|---|---|---|
| Quarterfield Elem. | 74.5 | 70.9 | −3.6 |
| Glen Burnie Park Elem. | 70.2 | 71.3 | +1.2 |
| Rippling Woods Elem. | 64.3 | 68.8 | +4.5 |
| Severn Elem. | 62.1 | 58.3 | −3.7 |
| Southgate Elem. | 59.9 | 63.1 | +3.2 |

The most-exposed school today, **Quarterfield (74.5)**, is among those helped most (−3.6, to 70.9), and Severn also falls. Two schools see real increases, stated plainly: **Rippling Woods** rises +4.5 (64.3 → 68.8) and **Southgate** +3.2. The robust, cross-option fact is about altitude over children, not per-school peak: **WARYN-BRGHT is the only option studied that never sends a pass below ~2,000 ft over any school, while a low turn-at-altitude drops a roughly 1,800-ft overflight onto the schools nearest the runway.**

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## 10. Common misconceptions, cleared up

### "TrueNoise's numbers are above the FAA's 65 limit."

No — these are **single-event peaks** (the loudest instant of one flyover), and 65 is a **DNL** (24-hour average) land-use contour. Different quantities, different scales. Our numbers cannot be compared to, or converted into, a DNL 65 figure, and should not be run through DNL-based annoyance formulas. (§2)

### "What altitude do they turn at?" — there isn't one number.

On 2,889 measured flights, the turn **begins around 1,800 ft** and **completes around 2,300 ft** over the community, climbing to the west. Aircraft are **lowest over the close-in neighborhoods and schools** and higher by the western end. Quoting one height for "the turn" — high or low — is what fuels the confusion. (§6)

**"The monitor's readings are the worst case."**

The opposite. The fixed monitor sits ~3.4 nm out, in the turn-*completion* zone, where its measured median peak is 70.5 dBA. Homes closer to the turn, where aircraft are lower, hear **louder** passes. The published receptor figures are a **conservative, low-end anchor** for much of the community, not a maximum. (§4, §6)

**"Spreading the flights out will fix it."**

Spreading (dispersion) moves noise; it does not lower it. Loudness is set by altitude, not by how tightly the tracks bunch. Fanning a *low* turn across more streets gives each point slightly fewer events but spreads a louder-per-event overflight onto more homes — and if the turn is forced lower to achieve it, every home under the wider band is louder (on the order of +6 dB for a ~1,300-ft drop). Only raising the turn makes each pass quieter for everyone beneath it. (§6)

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## 11. Equipment and its limits

Measurements come from an **IEC 61672 Class 2** sound-level monitor and a co-located airnoise.io device. Class 2 is the appropriate grade for this work for two reasons. First, the analysis is **comparative** — today's procedure against a proposed one at the same homes — so any fixed instrument offset applies equally to both and cancels in the difference. Second, the instrument has been checked against a **calibrated meter** and found within ~0.8 dB (§5), comfortably inside the ±7 dB screening band. This is a measured community screening, **not a certified Class 1 network and not an FAA AEDT model**; where a regulatory-grade number is required, these figures identify *where* to look, not the final certified value.

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## 12. Scope limits and honest caveats

Carry these with any use of the figures:

- **Metric:** peak dBA only — not DNL, not $L_{Aeq,16h}$. Do not convert to either.
- **The west-curve share is load-bearing.** The whole west-relief / southeast-increase split rests on today's 15R curving west for ~79% of departures. That share is measured here but not independently ground-truthed against a full operations record; state it as the assumption it is.
- **The single line is weakest where flights fan out** — between the turn and BOBYJ the measured tracks diverge, so the "today" centerline is least constrained there.
- **The climb is a median;** individual departures run higher or lower (part of the ±7 dB band). It is validated where the receptor measures it; far out it is less tightly anchored.
- **Idealized single lines** understate how many receptors real (dispersed) operations affect. This is a design-track screening comparison; precise dispersion is for FAA AEDT modeling.
- **WARYN-BRGHT is unflown** — projection, not observation: the design line draped with the measured 15R climb.

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